

10 Academy: Artificial Intelligence Mastery

Week 11 Challenge Document

Change Point Analysis and Statistical Modeling of Time Series Data

Detecting changes and associating causes on time series data

Date: 04 Feb - 10 Feb 2026

Overview

Business Need

You are a data scientist at **Birhan Energies**, a leading consultancy firm specialising in providing data-driven insights and strategic advice to stakeholders in the energy sector. With a mission to help clients navigate the complexities of the global energy market, Birhan Energies focuses on delivering actionable intelligence that supports decision-making processes for investors, policymakers, and energy companies.

You are tasked with analyzing how big political and economic events affect Brent oil prices. Understand how political decisions, conflicts in oil-producing areas, international sanctions, and OPEC policy changes affect the market.

The oil market is very unstable. This makes it hard for investors to make good decisions, manage risks, and maximize returns. Policymakers need detailed analysis to create strategies for economic stability and energy security. Energy companies need accurate price forecasts to plan operations, control costs, and secure supply chains.

Your objectives are to:

1. Identify key events that have significantly impacted Brent oil prices over the past decade.
2. Quantify how much these events affect price changes using statistical methods.
3. Provide clear, data-driven insights to guide investment strategies, policy development, and operational planning.

The main goal of this analysis is to study how important events affect Brent oil prices. This will focus on finding out how changes in oil prices are linked to big events like political decisions, conflicts in oil-producing regions, global economic sanctions, and changes in Organization of the Petroleum Exporting Countries (OPEC) policies. The aim is to provide clear insights that can help investors, analysts, and policymakers understand and react to these price changes better.

Data and Features

The [dataset](#) contains historical Brent oil prices with daily prices from May 20, 1987, to September 30, 2022.

Data fields

- **Date:** Represents the date of the recorded Brent oil price. Each entry is formatted as 'day-month-year' (e.g., 20-May-87). The dataset covers daily prices from May 20, 1987, to September 30, 2022.
- **Price:** This column represents the price of Brent oil on the corresponding date. The price is recorded in USD per barrel.

Learning Outcomes:

Skills:

- Change Point Analysis & Interpretation
- Statistical Reasoning
- Using PyMC - a standard Bayesian modelling package in Python
- Analytical Storytelling with Data

Knowledge:

- Probability distributions and choosing the relevant one for a given task
- Bayesian inference
- Monte Carlo Markov Chain
- Model comparison
- Policy analysis

Communication:

- Reporting to government bodies

Team

Tutors:

- Kerod
- Filimon
- Mahbubah

Communication & Support

- **Slack channel:** #all-week11
- **Office hours:** Mon–Fri, 08:00–15:00 UTC

Key Dates

- Challenge Introduction - 10:30 AM UTC on Wednesday, 04 Feb 2026
- Interim Solution - 8:00 PM UTC on Sunday, 08 Feb 2026
- Final Submission - 8:00 PM UTC on Tuesday, 10 Feb 2026

Deliverables and Tasks

Success in this challenge comes from making smart, strategic decisions about where to focus your effort.

Your priorities should be:

1. **Deep, well-explained analysis** using the core Bayesian Change Point model
2. **Master the concepts** of Bayesian inference and change point detection (PyMC)
3. **Build a functional dashboard** that delivers insights efficiently—quality of insights matters more than tech stack complexity
4. **Clear communication** through a compelling report and intuitive dashboard

Task 1: Laying the Foundation for Analysis

Objective: Define the data analysis workflow and develop a thorough understanding of the model and data.

Instructions:

1. Defining the Data Analysis Workflow:
 - a. **Outline your analysis steps:** Clearly document the steps and processes involved in analyzing the Brent oil prices data from data loading to insight generation.
 - b. **Research and compile event data:** Research major geopolitical events, OPEC decisions, and economic shocks relevant to the oil market. Compile a structured dataset (e.g., a CSV file) containing at least 10-15 key events with their approximate start dates.
 - c. **State assumptions and limitations:** Identify and document any assumptions and limitations of your analysis. Crucially, include a discussion on the difference between identifying a statistical correlation in time and proving causal impact.
 - d. **Determine communication channels:** Identify the main media channels and formats for communicating results to stakeholders.
2. Understanding the Model and Data:
 - a. **Review key references:** Read the main references related to the project to grasp the key concepts and models being used.
 - b. **Analyze time series properties:** Before modeling, investigate the Brent oil price data for key properties:
 - i. Trend analysis
 - ii. Stationarity testing
 - iii. Volatility patternsDiscuss how these properties inform your modeling choices.

- c. **Explain change point models:** Describe the purpose of change point models in the context of analyzing price fluctuations and how they help identify structural breaks in the data.
- d. **Describe expected outputs:** Explain the expected outputs of a change point analysis (e.g., dates of changes, new parameter values) and its limitations.

Deliverables:

- A 1-2 page document outlining your planned analysis steps
- A structured CSV/tabular data file with dates and event descriptions (minimum 10-15 events)
- Documentation of assumptions and limitations

Task 2: Change Point Modeling and Insight Generation

Objective: Apply Bayesian change point detection to identify and quantify structural breaks in Brent oil prices.

Instructions:

1. **Core Analysis (Mandatory)**
 - a. **Data Preparation and EDA:**
 - i. Load the data and convert the Date column to datetime format
 - ii. Plot the raw Price series over time to visually identify major trends, shocks, and periods of high volatility
 - iii. Consider analyzing log returns: $\log(\text{price}_t) - \log(\text{price}_{t-1})$ for stationarity
 - iv. Plot log returns to observe volatility clustering
 - b. **Build the Bayesian Change Point Model (in PyMC):**
 - i. **Define the Switch Point (τ):** Define it as a discrete uniform prior over all possible days in your dataset
 - ii. **Define "Before" and "After" Parameters:** For a simple model changing its mean, define two means (μ_1, μ_2)
 - iii. **Use a Switch Function:** Use `pm.math.switch` to select the correct parameter based on whether the time index is before or after τ
 - iv. **Define the Likelihood:** Connect the model to the data using a `pm.Normal` distribution where the mean is determined by the switch function
 - v. **Run the Sampler:** Use `pm.sample()` to run the MCMC simulation
 - c. **Interpret the Model Output:**
 - i. **Check for Convergence:** Use `pm.summary()` and look for $\hat{r}$ values close to 1.0. Examine trace plots using `pm.plot_trace()`

- ii. **Identify the Change Point:** Plot the posterior distribution of τ . A sharp, narrow peak indicates high certainty
 - iii. **Quantify the Impact:** Plot posterior distributions for before/after parameters and make probabilistic statements
- d. **Associate Changes with Causes:** Compare detected change point dates with your researched list of key events. Formulate hypotheses about which events likely triggered the shifts.
- e. **Quantify the Impact:** For each major change point associated with an event, describe the impact quantitatively. Example: *"Following the OPEC production cut announcement around [Date], the model detects a change point, with the average daily price shifting from \$X to \$Y, an increase of Z%."*

2. Advanced Extensions (Optional)

For a more advanced analysis, or to include in a "Future Work" section of your report, consider:

- **Explore Other Potential Factors:** Describe how you would incorporate other data sources (e.g., GDP, inflation rates, exchange rates) to build a more comprehensive explanatory model.
- **Consider Advanced Models:** Discuss how other models could provide different insights:
 - VAR (Vector Autoregression) for analyzing dynamic relationships between oil prices and macroeconomic variables
 - Markov-Switching models to explicitly define 'calm' vs. 'volatile' market regimes

Deliverables:

- Jupyter notebook with complete analysis code
- Visualizations of posterior distributions and change points
- Written interpretation of results with quantified impacts

Task 3: Developing an Interactive Dashboard for Data Analysis Results

Objective: Build a dashboard application to visualize analysis results and help stakeholders explore how various events affect Brent oil prices.

Instructions:

1. Backend (Flask)

- a. **Develop APIs** to serve data from your analysis results
- b. **Handle requests** for different datasets, model outputs, and performance metrics
- c. **Structure endpoints** for:
 - i. Historical price data

- ii. Change point results
- iii. Event correlation data

2. Frontend (React)

- a. **Create an intuitive interface** to display analysis results
- b. **Design interactive visualizations** showing how different events correlate with price changes
- c. **Include features:**
 - i. Filters and date range selectors
 - ii. Event highlight functionality to visualize price spikes/drops
 - iii. Drill-down capability for deeper insights
- d. **Ensure responsiveness** for desktop, tablet, and mobile devices
- **Recommended React Charting Libraries:**
 - [Recharts](#)
 - [React Chart.js 2](#)
 - [D3.js](#)

Key Dashboard Features

- Present historical trends and correlations with events (political decisions, conflicts, economic sanctions)
- Allow users to see how specific events influenced prices with "event highlight" functionality
- Display key indicators: volatility, average price changes around key events
- Enable data filtering and date range selection

Deliverables:

- Working Flask backend with documented API endpoints
- React frontend with interactive visualizations
- Screenshots demonstrating dashboard functionality
- README with setup instructions

Due Dates

Interim Submission — Sunday, 08 Feb 2026, 8:00 PM UTC

What to Submit:

- **GitHub Link** to your main branch covering Task 1
- **Interim Report** covering Task 1:
 - A 1-2 page document outlining your planned analysis steps
 - A structured tabular dataset with dates and event descriptions
 - Initial EDA findings

Feedback: You may not receive detailed comments on your interim submission, but you will receive a grade.

Final Submission — Tuesday, 10 Feb 2026, 8:00 PM UTC

What to Submit:

- **GitHub Link** to your main branch with complete code
- **Final Report** (blog post format suitable for Medium or PDF):
 - Complete analysis methodology and findings
 - Visualizations of change points and their associations with events
 - Quantified impact statements
 - Screenshots demonstrating dashboard functionality
 - Limitations and future work discussion

Feedback: You will receive comments/feedback in addition to a grade.

Other Considerations

- **Documentation:** Maintain detailed documentation in code comments and README files
- **Collaboration:** Use GitHub issues and projects for task tracking
- **Communication:** Regular check-ins, Q&A sessions, and a supportive community atmosphere
- **Flexibility:** Acknowledge potential challenges and communicate proactively
- **Professionalism:** Emphasize work ethics and professional behavior
- **Time Management:** Start early—Task 1 and Task 2 are foundational

Tutorials Schedule

In the following, the colour **purple** indicates morning sessions, and **blue** indicates afternoon sessions.

Wednesday: Data Analysis Techniques

- Introduction to the challenge (Kerod)
- Data Science Workflow (Filimon)

Thursday: Change Point Analysis

- Change Point Analysis (Mahbubah)
- Bayesian inference and Monte Carlo Markov Chain (Filimon)

Friday: Bayesian inference

- How to build React dashboards (Abdulhamid)
- Getting started with Flask (Mahbubah)

References

Data Science Workflow

1. <https://www.datascience-pm.com/data-science-workflow/>
2. <https://towardsdatascience.com/mastering-the-data-science-workflow-2a47d8b613c4>
3. <https://www.knowledgehut.com/blog/data-science/data-science-workflow>
4. <https://medium.com/codex/a-comprehensive-guide-to-master-the-data-science-workflow-739295117d67>

Change Point Analysis

1. <https://forecastegy.com/posts/change-point-detection-time-series-python/>
2. <https://eecs.wsu.edu/~cook/pubs/kais16.2.pdf>
3. <https://jagota-arun.medium.com/change-point-detection-in-time-series-bcf01409010e>
4. <https://www.iese.fraunhofer.de/blog/change-point-detection/>
5.  Change Point Detection in Time Series

Bayesian Change point detection

1. [Bayesian Changepoint Detection with PyMC3](#)
2. [Model Comparison using PyMC3](#)

Bayesian inference and Monte Carlo Markov Chain

1. [As a introduction to Bayesian statistics and the python package \(PyMC3\)](#)
2. https://warwick.ac.uk/fac/sci/statistics/staff/academic-research/steel/steels_homepage/bayesiantsrev.pdf
3. <https://www.embecosm.com/2021/12/18/forget-arima-going-bayesian-with-time-series-analysis/>
4. https://www.lancaster.ac.uk/~romano/post/change_point_online_intro/
5. https://www.pymc.io/blog/chris_F_pydata2022.html
6. <https://machinelearningmastery.com/markov-chain-monte-carlo-for-probability/>
7. <https://towardsdatascience.com/monte-carlo-markov-chain-mcmc-explained-94e3a6c8de11>

Alternative Data Source

1. [Petroleum Trade: Imports From OPEC Countries](#)
2. [Energy Consumption Transportation Sector](#)

React dashboard templates

1. <https://github.com/flatlogic/react-dashboard>

2. <https://github.com/creativetimofficial/light-bootstrap-dashboard-react?tab=readme-ov-file>
3. <https://github.com/flatlogic/react-dashboard>
4. <https://github.com/marmelab/react-admin>
5. <https://github.com/coreui/coreui-free-react-admin-template>

